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Department of Mathematics
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Math 201 Major Exam II Term 232

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The level surface of

$$f(x, y, z) = 4x^2 - y^2 + z^2$$

that passes through the point $(0, 1, 1)$ is

(a) a hyperboloid of two sheets

(b) a hyperboloid of one sheet

(c) an elliptic cone

(d) an elliptic paraboloid

(e) a hyperbolic paraboloid

2. An equation of the tangent plane to the surface $z = x^2 + y^2 + 3$ at the point $(2, 1, 8)$ is

(a) $3x + y - z = -1$

(b) $2x + 2y - z = -2$

(c) $4x - 2y - z = -2$

(d) $4x + 2y - z = 2$

(e) $4x + y + z = 17$

3. If $f(x, y, z) = \dfrac{z^2}{x + 2y}$, then $f_{xyz}(x, y, z) =$

(a) $\dfrac{2z}{(x + 2y)^2}$

(b) $-\dfrac{4z}{(x + 2y)^3}$

(c) $\dfrac{z}{(x + 2y)^3}$

(d) $\dfrac{2z + x}{(x + 2y)^2}$

(e) $\dfrac{8z}{(x + 2y)^3}$

4. Let $z = e^{xy}$, $x = \ln\left(\frac{s}{t}\right)$, $y = s^2 - t^2$. The value of $\frac{\partial z}{\partial t}$ when $s = 2$ and $t = 1$ is equal to

(a) $-8 - 8\ln 2$

(b) $16 - 16\ln 2$

(c) $16 - 8\ln 2$

(d) $-24 - 16\ln 2$

(e) $12 + 16\ln 2$

5. The direction of minimum increase of

$$f(x, y, z) = e^{2y}\cos(x + z^3)$$

at the point $\left(\frac{\pi}{2}, \frac{1}{2}, 0\right)$ is given by

(a) $-e\vec{i} + 2e\vec{j} + e\vec{k}$

(b) $-e\vec{i} + e\vec{k}$

(c) $e\vec{i}$

(d) $e\vec{j}$

(e) $e\vec{i} - e\vec{j}$

6. The distance between the point $(1, 3, -1)$ and the plane $3x - 4y + 5z = 6$ is equal to

(a) $4\sqrt{2}$

(b) $\frac{\sqrt{2}}{4}$

(c) $2\sqrt{2}$

(d) $6\sqrt{2}$

(e) $\frac{\sqrt{2}}{3}$

7. The domain D and the range R of the function

$$f(x, y) = 6 - 2x - 3y$$

are

(a) $D = \{(x, y) : -\infty < x < \infty,\ -\infty < y < \infty\}$, $R = (-\infty, 6)$

(b) $D = \{(x, y) : x \geq 0,\ y \geq 0\}$, $R = [0, \infty)$

(c) $D = \{(x, y) : -\infty < x < \infty,\ -\infty < y < \infty\}$, $R = (-\infty, \infty)$

(d) $D = \{(x, y) : x \geq 0,\ y \leq 0\}$, $R = (-\infty, 0)$

(e) $D = \{(x, y) : -\infty < x < \infty,\ -\infty < y < \infty\}$, $R = (6, \infty)$

8. The graph of the equation

$$4x^2 - 3y^2 + 12z^2 + 12 = 0$$

is

(a) an elliptic paraboloid

(b) an elliptic cone

(c) a hyperboloid of two sheets

(d) a hyperboloid of one sheet

(e) an ellipsoid

9. Using differentials, the quantity

$$(2.01)^2(9.02) - 2^2 \cdot 9$$

is approximately equal to

(a) 0.24

(b) 0.44

(c) 0.48

(d) 0.36

(e) 0.39

10. The maximum value of the directional derivative of

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

at the point $(1, 4, 2)$ is equal to

(a) 1

(b) 2

(c) $\frac{1}{4}$

(d) $\frac{1}{2}$

(e) $\frac{3}{2}$

11. If (a, b, c) is a point on the surface

$$z = x^2 - xy + y^2 - 2x - 2y$$

at which the tangent plane is horizontal, then $a^2 + b^2 + c^2 =$

(a) 24

(b) 28

(c) 20

(d) 32

(e) 36

12. If $f(x, y) = e^{-2y} \sin(\pi xy)$, then $f_y\left(1, \frac{1}{2}\right) =$

(a) $-\frac{2}{e}$

(b) $-\frac{\pi}{2}$

(c) $e\pi$

(d) π

(e) $\frac{1}{e}$

13. If $x \ln y + y^2 z + z^2 = 8$, then $\frac{\partial z}{\partial y} =$

(a) $-\frac{\ln y}{y^2 + 2z}$

(b) $-\frac{x + 2y^2 z}{y^2 + 2z}$

(c) $-\frac{x + 2y^2 z}{y^3 + 2yz}$

(d) $\frac{x - 2y^2 z}{y^3 - 2yz}$

(e) $\frac{x - 2y^2 z}{y^3 + 2yz}$

14. $\lim_{(x,y)\to(0,0)} \dfrac{2x-y^2}{x^3+y}$

(a) is equal to -1

(b) is equal to 1

(c) is equal to 0

(d) is equal to 2

(e) does not exist

15. The directional derivative of $f(x,y) = 3x - 4xy + 9y$ at $(1,2)$ in the direction of $\vec{v} = 3\vec{i} + 4\vec{j}$ is equal to

(a) 1

(b) $\dfrac{13}{5}$

(c) -8

(d) -1

(e) 3

16. $\lim_{(x,y)\to(0,0)} \dfrac{\sin(2x^2+2y^2)}{4x^2+4y^2} =$

(a) 1

(b) 2

(c) $\dfrac{1}{4}$

(d) 0

(e) $\dfrac{1}{2}$

17. Which one of the following statements is True about the function

$$f(x,y) = \begin{cases} \dfrac{x^4-y^4}{x^2+y^2}, & (x,y) \neq (0,0) \\ 1, & (x,y) = (0,0) \end{cases}$$

(a) f is not continuous at $(0,0)$

(b) $\lim_{(x,y)\to(0,0)} f(x,y) = -1$

(c) f is continuous at $(0,0)$

(d) $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist

(e) $\lim_{(x,y)\to(0,0)} f(x,y) = 1$

18. If (a, b, c) is a point on the ellipsoid $3x^2 + y^2 + 3z^2 = 1$ where the tangent plane is parallel to the plane $-6x + y + 3z = 0$, then $a + b + c =$

(a) $\frac{1}{2}$

(b) $-\frac{1}{4}$

(c) 1

(d) -1

(e) 0

19. If $ax + by + cz = 4$ is an equation for the plane that passes through the point $(2, 2, 1)$ and contains the line $x = y - 4 = z$, then $abc =$

(a) 3

(b) 10

(c) -12

(d) -4

(e) -15

20. A set of parametric equations for the line passing through the points $P(2, 0, 1)$ and $Q(1, -1, 2)$ is given by

(a) $x = 2 - 2t,\ y = 1 - t,\ z = 1 + t$

(b) $x = -t,\ y = 2 - t,\ z = t$

(c) $x = 2 - t,\ y = -t,\ z = 1 + t$

(d) $x = 2 - t,\ y = -2t,\ z = 1 + t$

(e) $x = 2 + t,\ y = -t,\ z = 1 - t$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$c = f(0,1,1) = 0$, so $y^2 = 4x^2 + z^2$: an elliptic cone.
2	**(d)**	$f_x = 4$, $f_y = 2$: $4(x-2) + 2(y-1) - (z-8) = 0 \Rightarrow 4x + 2y - z = 2$.
3	**(e)**	$f_x = -z^2(x+2y)^{-2}$, $f_{xy} = 4z^2(x+2y)^{-3}$, $f_{xyz} = \frac{8z}{(x+2y)^3}$.
4	**(d)**	Chain rule at $(2,1)$: $24(-1) + 8\ln 2\,(-2) = -24 - 16\ln 2$.
5	**(c)**	$\nabla f = \langle -e, 0, 0\rangle$, so the direction of minimum increase is $-\nabla f = e\vec{i}$.
6	**(c)**	$d = \frac{\lvert 3-12-5-6\rvert}{\sqrt{50}} = \frac{20}{5\sqrt{2}} = 2\sqrt{2}$.
7	**(c)**	Polynomial: $D = \mathbb{R}^2$; $f(x,0) = 6 - 2x$ covers all reals, so $R = (-\infty, \infty)$.
8	**(c)**	$-\frac{x^2}{3} + \frac{y^2}{4} - z^2 = 1$: two minus signs, hyperboloid of two sheets.
9	**(b)**	$f = x^2y$: $df = 2xy\,dx + x^2dy = 36(0.01) + 4(0.02) = 0.44$.
10	**(a)**	$\nabla f(1,4,2) = \frac{1}{\sqrt{21}}\langle 1,4,2\rangle$, so the maximum is $\lVert\nabla f\rVert = 1$.
11	**(a)**	$f_x = f_y = 0$ gives $(2,2,-4)$, so $a^2 + b^2 + c^2 = 4 + 4 + 16 = 24$.
12	**(a)**	$f_y = -2e^{-2y}\sin(\pi xy) + \pi x e^{-2y}\cos(\pi xy)$, at $(1, \frac{1}{2})$ equals $-\frac{2}{e}$.
13	**(c)**	$F = x\ln y + y^2z + z^2 - 8$: $-\frac{F_y}{F_z} = -\frac{x/y + 2yz}{y^2 + 2z} = -\frac{x + 2y^2z}{y^3 + 2yz}$.
14	**(e)**	Along $x = 0$ the limit is 0, along $y = x$ it is 2: does not exist.
15	**(a)**	$\nabla f(1,2) = \langle -5, 5\rangle$, $\mathbf{u} = \langle \frac{3}{5}, \frac{4}{5}\rangle$: $D_{\mathbf{u}}f = 1$.
16	**(e)**	With $u = 2x^2 + 2y^2$: $\frac{\sin u}{2u} \to \frac{1}{2}$.
17	**(a)**	Limit is $\lim(x^2 - y^2) = 0 \neq f(0,0) = 1$, so f is not continuous at $(0,0)$.
18	**(e)**	$\nabla F \parallel \langle -6, 1, 3\rangle$ gives $(a,b,c) = \pm(-\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, so $a + b + c = 0$.
19	**(c)**	Normal $\langle -3, -1, 4\rangle$ gives $3x + y - 4z = 4$, so $abc = 3 \cdot 1 \cdot (-4) = -12$.
20	**(c)**	$\overrightarrow{PQ} = \langle -1, -1, 1\rangle$ from $P(2,0,1)$: $x = 2 - t,\ y = -t,\ z = 1 + t$.